Dhia Eddine Arfaoui

Machine Learning — Deep Learning — GenAI

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Professional Summary

Machine Learning and AI engineer with hands-on experience developing forecasting and classification models, data pipelines, and Python APIs across banking, insurance, and toll operations. Builds deep-learning and generative-AI applications using TensorFlow, PyTorch, and locally hosted language models. Currently pursuing a Master's in Artificial Intelligence, Data and Agentic Systems, with a focus on ML, deep learning, and agentic AI opportunities.

Experience

Data Science Intern *February 2026 – July 2026*
Numeryx *Ariana, Tunisia*

- Developed an end-to-end traffic and revenue forecasting solution for the TollXpress toll network, integrating ML models with a PostgreSQL warehouse, REST API, and dashboard.
- Benchmarked ARIMA, Prophet, and LSTM for traffic forecasting, and Linear Regression, Random Forest, and XGBoost for revenue forecasting; reported model precision above 85%.
- Built ETL workflows and a star-schema warehouse consolidating toll transactions from lanes, stations, and the central system to support modeling and analysis.
- Technologies: Python, Pandas, NumPy, Scikit-learn, TensorFlow, FastAPI, PostgreSQL, Docker, Git.

Data and ML Intern *Jun 2025 – Jul 2025*
BIAT *Tunis, Tunisia*

- Developed an internal API to analyze daily banking transactions and surface quantitative and qualitative indicators.
- Built SQL and Oracle pipelines to process daily financial transaction records for reporting and downstream analytics.
- Performed exploratory data analysis and feature engineering to prepare data for analytics and modeling.
- Technologies: Python, Flask, SQL, Oracle, TensorFlow, Keras, Docker,Git.

Data Science Intern *June 2024 – July 2024*
BIAT Assurances *Tunis, Tunisia*

- Developed an automated pipeline for stochastic analysis of bonus-malus insurance classes.
- Prepared and structured insurance datasets for statistical analysis and feature engineering.
- Technologies: Python, Flask, numpy , Scikit learn , Git.

Education

Master in Artificial Intelligence, Data and Agentic (AIDA) *2026 – Present*
Université Paris Dauphine–PSL, Tunis Campus

- Focus areas: artificial intelligence, machine learning, deep learning, Big Data, and agentic and conversational systems.
- Applied work in AI model development, algorithms, cloud platforms, and Big Data tools.

Engineering Degree in Industrial Engineering *2023 – 2026*
ENSIT *Montfleury, Tunisia*

- Engineering studies spanning optimization, data engineering, quantitative finance, and software development with C and Java.

Technical Skills

- **Programming:** Python, SQL, Java, C, JavaScript
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
- **Generative & Agentic AI:** Ollama, LangChain, LangGraph, local LLM applications , Hugging face
- **NLP & Retrieval:** NLTK, Gensim, spaCy, FAISS
- **Data :** Pandas, NumPy, PostgreSQL, Oracle, SQLite, Hadoop, Spark, HDFS, MapReduce , Big querry
- **APIs & Web:** FastAPI, Flask, SQLAlchemy, Streamlit, React, Spring Boot
- **deployement and tools :** Git , Docker , Google cloud platform

Selected Projects

NLP Text Generation (Encoder–Decoder) *NLP, Neural Networks, Sequence Generation*

- Developing an end-to-end neural text-generation pipeline covering text preprocessing, tokenization, vocabulary construction, embeddings, encoder–decoder modeling, training, and generation.
- Evaluating generated text with quantitative and qualitative analysis, baseline comparisons, and review of model strengths and limitations.

AI Document Q&A *FastAPI, Python, PostgreSQL, SQLAlchemy, Ollama*

- Built a browser-based app for natural-language questions about uploaded TXT, PDF, and DOCX files; extracted and chunked content, then indexed document embeddings in a vector database for similarity retrieval.
- Retrieved relevant context for a locally hosted Ollama model to generate grounded answers without relying on paid API keys.

Data Report Analyzer *Python, FastAPI, Pandas, SQLAlchemy, SQLite, LangChain, Ollama*

- Built a CSV-to-report workflow that identifies column roles, profiles dataset fields, and calculates relevant statistics.
- Used a locally hosted Ollama model to generate a narrative from the analysis, then stored the dataset and report in a database.

Certifications

- Agentic AI Certified Foundations Associate, Oracle (Issued Oct 10, 2026; valid through Oct 10, 2028; Credential ID: 103558704AAI26OFA)
- Associate Data Scientist, DataCamp (Issued Nov 2025, Credential ID: DSA0017594101523)
- Data Science Certification, CodeQuest (Apr 2025)